GG-S-161/3D

May 1, 1985

SUPERSEDING
GG-S-161/3C
May 3, 1977

FEDERAL SPECIFICATION SHEET

SCALES, DRAFTING: MECHANICAL

This specification was approved by the Assistant Administrator, Officer of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring Scales, Drafting: Mechanical described herein shall consist of this document and the latest issue of GG-S-161.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The specification covers detail requirements for scales graduated in fractions of an inch to the inch, commonly called "size scales", and size scales graduated in the metric system.

1.2 Classification.

1.2.1 Type designation number. Scales covered by this specification shall be identified by a type designation number (see 6.2) formed as indicated in GG-S-161. The type designation number is intended for cataloging and ordering purposes and not for printing on the scales. Sub-paragraphs 1.2.1.3 through 1.2.1.10 indicate the options available for the type III scale and are further identified in GG-S-161.

1.2.1.1 Spec. No. GG-S-161.

1.2.1.2 Type of scales - III.

1.2.1.3 Shape of scales - -1, -2, -3, or -4.

1.2.1.4 Composition of scales - -A, -B, -C, -D, -E, -F, or -G.

1.2.1.5 Grade of scales - -1, or -2.

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1.2.1.6 Size of scales - -B or -F.

1.2.1.7 Scale graduations - -1.

1.2.1.8 Numbering of scales - -A.

1.2.1.9 Style of scales - -A, -B, -C, -D, -E, -F, -G, -H, -J, -K, -L, or -M (see 3.2).

1.2.1.10 Type of sheath required - -1, -2, -3, -4, -5, or -6.

3. REQUIREMENTS

3.1 Description. Scales shall be in accordance with GG-S-161 and as specified herein.

3.2 Shapes and styles.

3.2.1 Shapes 1, 2, and 3. Shapes 1, 2, and 3 shall be as shown in figure 1 of GG-S-161, and shall be direct numbered and fully divided in the scale divisions specified in the following styles (see 1.2.1.9).

3.2.1.1 Style A. The style A scale shall have one edge scaled to full-size (12 inches per foot) reading to thirty-seconds, numbered at not less than each inch. The other edge shall be scaled to half-size (6 inches per foot) reading to sixteenth, numbered at not less than each inch.

3.2.1.2 Style B. The style B scale shall have one edge scaled to full-size (12 inches per foot) reading to thirty-seconds, numbered at not less than each inch. The other edge shall be scaled to quarter-size (3 inches per foot) reading to eighths, numbered at not less than every other inch.

3.2.1.3 Style C. The style C scale shall have one edge scaled to full-size (12 inches per foot) reading to thirty-seconds, numbered at not less than each inch. The other edge shall be scaled eighth-size (1.5 inches per foot) reading to fourths, numbered at not less than every other inch.

3.2.1.4 Style D. The style D scale shall have one edge scaled to half-size (6 inches per foot) reading to sixteenths, numbered at not less than each inch. The other edge shall be scaled to quarter-size (3 inches per foot) reading to eighths, numbered at not less than every other inch.

3.2.1.5 Style E. The style E scale shall have one edge scaled to half-size (6 inches per foot) reading to sixteenths, numbered at not less than each inch. The other edge shall be scaled to eighth-size (1.5 inches per foot) reading to fourths, numbered at not less than every other inch.

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3.2.1.6 Style F. The style F scale shall have one edge scaled to quarter-size (3 inches per foot) reading to eighths, numbered at not less than every other inch. The other edge shall be scaled to eighth-size (1.5 inches per foot) reading to fourths, numbered not less than every other inch.

3.2.1.7 Style G. The style G scale shall have one edge scaled to full-size (100 centimeters per meter) reading to 1 millimeter (mm), numbered at not less than every centimeter. The other edge shall be scaled to half-size (50 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter.

3.2.1.8 Style H. The style H scale shall have one edge scaled to full-size (100 centimeters per meter) reading to 1 mm, numbered at not less than every centimeter. The other edge shall be scaled to quarter-size (25 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter.

3.2.1.9 Style J. The style J scale shall have one edge scaled to full-size (100 centimeters per meter) reading to 1 mm, numbered at not less than every centimeter. The other edge shall be scaled to eighth-size (12.5 centimeters per meter) reading to 1 mm, numbered at not less than every fourth centimeter.

3.2.1.10 Style K. The style K scale shall have one edge scaled to half-size (50 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter. The other edge shall be scaled to quarter-size (25 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter.

3.2.1.11 Style L. The style L scale shall have one edge scaled to half-size (50 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter. The other edge shall be scaled to eighth-size (12.5 centimeters per meter) reading to 1 mm, numbered at not less than every fourth centimeter.

3.2.1.12 Style M. The style M scale shall have one edge scaled to quarter-size (25 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter. The other edge shall be scaled to eighth-size (12.5 centimeters per meter) reading to 1 mm, numbered at not less than every fourth centimeter.

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3.2.2 Shape 4. Shape 4 shall be as shown in figure 1 of GG-S-161 and shall be direct numbered and fully divided in the scale divisions specified in the following styles (see 1.2.1.9).

3.2.2.1 Style A. The style A scale shall have one edge scaled to full-size (12 inches per foot) reading to thirty-seconds, numbered at not less than each inch. The second edge shall be scaled to half-size (6 inches per foot) reading to sixteenths, numbered at not less than each inch. The third edge shall be scaled to quarter-size (3 inches per foot) reading to eighths, numbered at not less than every other inch. The fourth edge shall be scaled to eighth-size (1.5 inches per foot) reading to fourths, numbered at not less than every other inch.

3.2.2.2 Style B. The style B scale shall have one edge scaled to full-size (100 centimeters per meter) reading to 1 mm, numbered at not less than every centimeters per meter) reading to 1 mm, numbered at not less than every centimeter. The second edge shall be scaled to half-size (50 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter. The third edge shall be scaled to quarter-size (25 centimeters per meter) reading to 1 mm, numbered at not less than every other centimeter. The fourth edge shall be scaled to eighth-size (12.5 centimeters per meter) reading to 1 mm, numbered at not less than every fourth centimeter.

3.3 Scale length. Scale length for shapes 1, 2, and 3 shall be 12 inches/30 centimeters nominal. For the shape 4, scale length shall be 6 inches/15 centimeters nominal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with GG-S-161.

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be in accordance with GG-S-161.

6. NOTES. (The notes in GG-S-161 are applicable to this specification.)

6.1 Intended use. Scales are used in mechanical drafting.

6.2 Ordering data. Procurement documents should specify those desired options offered herein and should also specify the following:

a. Type designation number (see 1.2.1).

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MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review activities

Army - GL
Navy - MC
DLA - GS

User activity

Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITIES:

FSS
NBS
OSS
SCS

PREPARING ACTIVITY:

Army - ME

Project 6675-0783

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